

Scenario: Management of children and young people with type 1 diabetes

Last revised in December 2024

From age 12 months to 17 years.

What initial care and support should I offer a child or young person with type 1 diabetes?

- Ensure that children and young people with type 1 diabetes are offered an ongoing integrated package of care (</topics/diabetes-type-1/management/management-children-young-people/#care-plan>), provided by a multidisciplinary paediatric diabetes team.
 - Ensure that the child or young person and/or their family/carers are involved in making decisions about their care.
- Ensure that they are offered a continuing programme of education.
 - This should be tailored to the child/young person and include some core elements (</topics/diabetes-type-1/management/management-children-young-people/#care-plan>).
- Encourage children and young people with type 1 diabetes and/or their family/carers to:
 - Have the following immunizations recommended by the Department of Health (if applicable):
 - Annual immunization against influenza (if aged over 6 months).
 - Immunization against pneumococcal infection.
 - Wear or carry something that identifies them as having type 1 diabetes, for example:
 - A MedicAlert® bracelet, necklace, or watch (available from www.medicalert.org.uk (<http://www.medicalert.org.uk/>)).
 - Diabetes ID Card (available from www.diabetes.co.uk (<https://www.diabetes.co.uk/diabetic-products/medipal-card.html>)).
 - Discuss any concerns or questions with their diabetes specialist team.
- Provide information and advice on:
 - Means and frequency of communicating with the paediatric diabetes team.
 - Encourage children and young people with type 1 diabetes and/or their family/carers to attend clinic 4 times a year as regular contact is associated with optimal blood glucose control.
 - Ensure that the child or young person and/or their family/carers know how to contact the diabetes team during working hours and out of hours, as available.
 - Government disability benefits — Benefits calculators (<https://www.gov.uk/benefits-calculators>) are available at www.gov.uk (<https://www.gov.uk/>).
- Manage lifestyle issues, such as diet, exercise, and smoking. See the section on Lifestyle management (</topics/diabetes-type-1/management/management-children-young-people/#lifestyle-management>) for further information.

- **Provide up-to-date information on diabetes support groups (local and national),** including information on how to contact them and the benefits of membership.
 - For example, the Diabetes UK website (www.diabetes.org.uk (<https://www.diabetes.org.uk/>)) has lots of information on [Living with diabetes](https://www.diabetes.org.uk/guide-to-diabetes) (<https://www.diabetes.org.uk/guide-to-diabetes>) and a [support helpline](https://www.diabetes.org.uk/how_we_help/helpline) (https://www.diabetes.org.uk/how_we_help/helpline) for people with diabetes and their family/carers.
 - Provide this information at the time of diagnosis and periodically thereafter. Written information should also be available from the diabetes team.

Care plan for children and young people with type 1 diabetes

- **At the time of diagnosis, children and young people with type 1 diabetes and/or their family/carers should be offered an integrated package of care provided by a multidisciplinary paediatric diabetes team.**
 - To optimize the effectiveness of care and reduce the risk of complications, the diabetes team should include members with appropriate training in clinical, educational, dietetic, lifestyle, mental health, and foot care aspects of diabetes for children and young people.
 - The child or young person may be offered home-based or inpatient management, depending on clinical need and family circumstances and wishes.
 - Home-based care with support from the local paediatric diabetes team (including 24-hour telephone access) is safe and as effective as inpatient initial management.
 - Initial inpatient management may be offered to children younger than 2 years of age; children and young people with social or emotional factors that would make home-based management inappropriate; and children and young people who live far away from the hospital.
- **They should also be offered a continuing programme of education.** This should:
 - Include the following core elements:
 - Insulin therapy, including its aims, how it works, its mode of delivery, and dosage adjustment.
 - Blood glucose monitoring, including targets for blood glucose control (blood glucose and HbA1c levels).
 - The effects of diet, physical activity, and intercurrent illness on blood glucose control.
 - Managing intercurrent illness ('sick-day rules', including monitoring of blood ketones).
 - Detecting and managing hypoglycaemia, hyperglycaemia, and ketosis.
 - Be tailored to the child/young person and/or their family/carers, taking account of issues such as:
 - Emotional wellbeing.
 - Age and maturity.
 - Cultural considerations.
 - Existing knowledge.
 - Current and future social circumstances.
 - Life goals.

Basis for recommendation

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Insulin therapy for children and young people with type 1 diabetes

- Insulin therapy should only be initiated and managed by healthcare professionals with the relevant expertise and training.
- See the CKS topic on [Insulin therapy in type 1 diabetes \(/topics/insulin-therapy-in-type-1-diabetes/\)](/topics/insulin-therapy-in-type-1-diabetes/) for detailed information on insulin therapy, including available insulin preparations, recommended regimens, information on injection technique and injection sites, storage of insulin, adverse effects, safety issues, and general information on lifestyle and cultural aspects of insulin treatment (such as driving whilst on insulin treatment, fasting, shift work, holiday and travel, and insurance).

What is the target HbA1c for children and young people with type 1 diabetes?

- Explain to children and young people with type 1 diabetes and/or their family/carers that an HbA1c target level of 48 mmol/mol (6.5%) or lower is ideal to minimize the risk of long-term [complications \(/topics/diabetes-type-1/background-information/complications/\)](/topics/diabetes-type-1/background-information/complications/) of type 1 diabetes.
 - Explain the benefits of safely achieving and maintaining the lowest attainable HbA1c.
 - If the child or young person has an HbA1c level above the ideal target, explain that any reduction in HbA1c level reduces the risk of long-term complications.
 - Agree an individualized lowest achievable HbA1c target with each child or young person with type 1 diabetes and/or their family/carers, taking into account factors such as daily activities, individual life goals, complications, comorbidities, and the risk of hypoglycaemia (generally defined as blood glucose levels less than 3.5 mmol/L).
 - Support them to safely achieve and maintain their individually agreed HbA1c target level.
 - Ensure that children and young people with type 1 diabetes are offered measurement of their HbA1c level 4 times a year. More frequent testing may be appropriate if there is concern about suboptimal blood glucose control.

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What should I advise on self-monitoring of blood glucose?

- Self-monitoring of blood glucose refers to home blood glucose testing for people with diabetes to determine the amount of glucose in the blood at any given time.
- Continuous glucose monitoring is a way of measuring glucose levels continuously throughout the day and night via a tiny electrode inserted under the skin.
 - Continuous glucose monitoring systems provide continuous measurements (every 1–5 minutes) of glucose concentrations in the interstitial fluid, which correlate with blood glucose levels. It can provide both retrospective and real-time information to detect hypoglycaemic and hyperglycaemic episodes, predict impending hypoglycaemia, and detect wide fluctuations in glucose levels (glycaemic variability).
 - All children and young people with type 1 diabetes should be offered real-time continuous glucose monitoring (rtCGM) — intermittently scanned continuous glucose monitoring (isCGM) should be offered to children and young people with type 1 diabetes aged 4 years and over who are unable to use rtCGM or who express a clear

preference for isCGM. If the person cannot use or does not want rtCGM or isCGM, capillary blood glucose monitoring should be offered.

- CGM should be provided by a team with expertise in its use, as part of supporting people to self-manage their diabetes.
- Advise children and young people with type 1 diabetes who are using CGM (and their families and carers) that they will still need to take capillary blood glucose measurements (although they can do this less often). Explain that this is because:
 - They will need to use capillary blood glucose measurements to check the accuracy of their CGM device
 - They will need capillary blood glucose monitoring as a back-up (for example, when their blood glucose levels are changing quickly or if the device stops working).
- The child or young person's use of CGM should be monitored and reviewed as part of reviewing their diabetes care plan.
- All people with type 1 diabetes will need a blood glucose monitor, lancets (which fit into a finger-pricking device), and testing strips (on which the blood is collected for testing in the machine) to monitor capillary blood glucose.
 - See the section on [Target glucose self-monitoring levels \(/topics/diabetes-type-1/management/management-children-young-people/#target-glucose-self-monitoring-levels\)](#) for information on blood glucose self-monitoring target levels.
- **Self-monitoring skills should be taught at the time of diagnosis and at the start of insulin therapy, and should be reviewed at least annually.** The review should include checking that the person knows:
 - How to use their blood glucose monitoring meter — this can be done by asking them to demonstrate their technique.
 - When to test — for example before breakfast and 2 hours after meals, during periods of illness, before driving, and if they feel hypoglycaemic.
 - How to assess and respond to test results, including how to:
 - Interpret trends in blood glucose levels as well as individual readings — this is important for adjusting insulin doses.
 - Identify the causes of high and low blood glucose values.

What are the target blood glucose self-monitoring levels for children and young people with type 1 diabetes?

- **Advise children and young people who are using capillary blood glucose monitoring to monitor their blood glucose levels to routinely perform at least 5 capillary blood glucose tests per day.**
 - Advise the child or young people and/or their family/carers that more frequent testing is often needed (for example with physical activity and during intercurrent illness), and ensure they have enough test strips for this.
 - Offer a choice of equipment for monitoring capillary blood glucose so that they can optimize their blood glucose control in response to adjustment of insulin, diet, and exercise.
- **The optimal targets for glucose self-monitoring in children and young people with type 1 diabetes are:**
 - Fasting plasma glucose level of 4–7 mmol/L on waking.
 - Plasma glucose level of 4–7 mmol/L before meals at other times of the day.
 - Plasma glucose level of 5–9 mmol/L after meals.
 - For young people of driving age, plasma glucose level of at least 5 mmol/L when driving.

- Explain that achieving and maintaining blood glucose levels towards the lower end of the target optimal ranges will help them to achieve the lowest attainable HbA1c.
 - Ensure that children and young people do not experience problematic hypoglycaemia or undue emotional distress when achieving, or attempting to achieve, blood glucose targets.
 - Be aware that there may be conflict between the child or young person and their family members or carers about blood glucose targets and that an agreed compromise may be needed.
 - Explain that blood glucose levels should be interpreted in the context of the 'whole child', which includes the social, emotional, and physical environment.

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How should I manage lifestyle issues in children and young people with type 1 diabetes?

- See the sections on [Diet \(/topics/diabetes-type-1/management/management-children-young-people/#diet\)](#), [Exercise \(/topics/diabetes-type-1/management/management-children-young-people/#exercise-physical-activity\)](#), [Alcohol \(/topics/diabetes-type-1/management/management-children-young-people/#alcohol-intake\)](#), and [Smoking and substance misuse \(/topics/diabetes-type-1/management/management-children-young-people/#smoking-substance-misuse\)](#) for information on managing these lifestyle issues.
- Detailed information on managing other lifestyle and cultural issues, such as driving, fasting, holiday and travel, and shift work, are covered in the section on [Additional information and advice \(/topics/insulin-therapy-in-type-1-diabetes/management/insulin-therapy-type-1-diabetes/#lifestyle-cultural-issues\)](#) in the CKS topic on [Insulin therapy in type 1 diabetes \(/topics/insulin-therapy-in-type-1-diabetes/\)](#).

Advice on diet

- Support children and young people with type 1 diabetes and/or their family/carers to develop a good working knowledge of nutrition and how it affects their diabetes.
 - Explain how healthy eating, including eating foods with a low glycaemic index (GI), at least five portions of fruit and vegetables a day, and appropriate types and amounts of fats, can reduce their risk of cardiovascular disease (CVD).
 - Explain that a low GI diet may also help to improve blood glucose control and reduce the risk of hyperglycaemic episodes.
 - Support the child or young person and their family/carers to adjust their food choices accordingly, or consider referring them to a paediatric dietitian.
 - Offer level 3 carbohydrate-counting education (carbohydrate counting with adjustment of insulin dosage according to an insulin:carbohydrate ratio) to children and young people who are using a multiple daily insulin injection regimen or continuous subcutaneous insulin infusion (CSII or insulin pump) therapy. Repeat the offer at intervals thereafter.
 - See the section on [Insulin regimens \(/topics/insulin-therapy-in-type-1-diabetes/management/insulin-therapy-type-1-diabetes/#recommended-insulin-regimen\)](#) in the CKS topic on [Insulin therapy in type 1 diabetes \(/topics/insulin-therapy-in-type-1-diabetes/\)](#) for information on insulin treatment regimens.

- Explain that children and young people with type 1 diabetes have the same basic nutritional needs as other children and young people so their food should provide sufficient energy and nutrients for optimal growth and development.
- If necessary, offer the child or young person dietetic support to help optimize body weight.
- **Take into account social and cultural considerations when providing advice on dietary management to children and young people with type 1 diabetes.**

Advice on exercise and physical activity

- **Advise children and young people with type 1 diabetes and/or their family/carers that regular exercise may lower glucose levels, as exercise increases the amount of glucose used by the muscles for energy.**
 - Explain that:
 - They can take part in all forms of exercise, provided appropriate attention is given to diet and changes in insulin.
 - Changes in daily exercise patterns may require insulin dose and/or carbohydrate intake to be altered.
 - Offer advice on the effects of exercise on blood glucose levels, and advise on strategies for avoiding hypo- or hyperglycaemia during or after physical activity. For example:
 - Additional carbohydrate should be consumed as appropriate to avoid hypoglycaemia.
 - Carbohydrate-based foods should be readily available during and after exercise.
 - Additional carbohydrate should be consumed if plasma glucose levels are less than 7 mmol/L before exercise is undertaken.
 - See the section on [Managing hypoglycaemia \(/topics/insulin-therapy-in-type-1-diabetes/management/insulin-therapy-type-1-diabetes/#managing-an-episode-of-hypoglycaemia\)](#) in the CKS topic on [Insulin therapy in type 1 diabetes \(/topics/insulin-therapy-in-type-1-diabetes/\)](#) for information on managing a hypoglycaemic episode.
 - Encourage them to monitor blood glucose levels before and after exercise so that they can:
 - Identify when changes in insulin or food intake are necessary.
 - Learn the blood glucose response to different exercise conditions.
 - Be aware of exercise-induced hypoglycaemia (which may occur several hours after prolonged exercise).
- **Also advise that:**
 - Regular exercise can reduce the risk of cardiovascular disease in the long term.
 - Exercise can help with weight management (when combined with a healthy diet). See the CKS topic on [Obesity \(/topics/obesity/\)](#) for more information.
 - They should seek advice from their diabetes team if they wish to participate in a sport that may have particular risks for people with diabetes. Additional information may be available from local and/or national support groups and organizations, such as Diabetes UK (www.diabetes.org.uk (<https://www.diabetes.org.uk/>)).
- **The current [UK Chief Medical Officers' Physical Activity Guidelines \(/assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/832868/uk-chief-medical-officers-physical-activity-guidelines.pdf\)](#) state that:**
 - **Toddlers (1–2 years) should:**

- Spend at least 180 minutes per day in a variety of physical activities at any intensity, including active and outdoor play, spread throughout the day.
- **Children aged 3–4 years should:**
 - Spend at least 180 minutes per day in a variety of physical activities spread throughout the day, including active and outdoor play. The 180 minutes should include at least 60 minutes of moderate-to-vigorous intensity physical activity.
- **Children and young people aged 5–18 years should:**
 - Engage in moderate-to-vigorous intensity physical activity for an average of at least 60 minutes per day across the week. This can include all forms of activity, such as physical education, active travel, after-school activities, play, and sports.
 - Engage in a variety of types and intensities of physical activity across the week to develop movement skills, muscular fitness, and bone strength.
 - Aim to minimize the amount of time spent being sedentary. When physically possible, they should break up long periods of not moving with at least light physical activity.

Advice on alcohol intake

- **Remind young people and/or their family/carers that the legal drinking age in the UK is 18 years.**
- **Advise young people with type 1 diabetes who drink:**
 - To avoid drinking alcohol on an empty stomach as the alcohol will be absorbed faster. They should:
 - Eat a snack that contains carbohydrate (such as a sandwich or crisps) before and after drinking alcohol. Extra insulin is not required.
 - Measure their blood glucose regularly, and maintain their blood glucose with appropriate carbohydrate intake.
 - That alcohol may exacerbate or prolong the hypoglycaemic effect of insulin, including nocturnal hypoglycaemia.
 - Signs of hypoglycaemia may become less clear, and delayed hypoglycaemia may occur (which can be hours after alcohol consumption).
 - See the section on [Hypoglycaemia \(/topics/insulin-therapy-in-type-1-diabetes/management/insulin-therapy-type-1-diabetes/#managing-hypoglycaemia\)](#) in the CKS topic on [Insulin therapy in type 1 diabetes \(/topics/insulin-therapy-in-type-1-diabetes/\)](#) for detailed information on hypoglycaemia.
 - To always wear some form of diabetes identification (as the reduced awareness of hypoglycemia may be confused with alcohol intoxication). Examples of diabetes identification include:
 - A MedicAlert® bracelet, necklace, or watch (available from www.medicalert.org.uk (<http://www.medicalert.org.uk/>)).
 - Diabetes ID Card (available from www.diabetes.co.uk (<https://www.diabetes.co.uk/diabetic-products/medipal-card.html>)).
- **Give information on the recommended maximum alcohol intake.**
 - The [Chief Medical Officers' guideline](#) (http://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/545739/GDG_report-Jan2016.pdf) for both men and women states that:
 - It is safest not to regularly drink more than 14 units of alcohol per week, to keep health risks from drinking alcohol to a low level.
 - People who drink as much as 14 units of alcohol per week should spread this evenly over 3 days or more. One or two heavy drinking sessions increases

the risk of death from long-term illnesses and from accidents and injuries.

- The risk of developing a range of illnesses (such as cancers of the mouth, throat, and breast) increases with any amount of alcohol consumed on a regular basis.
- If people wish to cut down the amount of alcohol they are drinking, a good way to help achieve this is to have several drink-free days each week.
- For women who are pregnant or planning a pregnancy, the safest approach is not to drink alcohol at all, to keep risks to the baby to a minimum.
- See the section on [Unit of alcohol \(/topics/alcohol-problem-drinking/background-information/definition/#unit-of-alcohol/\)](#) in the CKS topic on [Alcohol – problem drinking \(/topics/alcohol-problem-drinking/\)](#) for more information on alcohol units.
- Information on [Alcohol and diabetes \(https://www.diabetes.org.uk/Guide-to-diabetes/Enjoy-food/what-to-drink-with-diabetes/Alcohol-and-diabetes\)](https://www.diabetes.org.uk/Guide-to-diabetes/Enjoy-food/what-to-drink-with-diabetes/Alcohol-and-diabetes) and [Type 1 diabetes and drinking \(https://www.diabetes.org.uk/guide-to-diabetes/young-adults/type-1-drinking\)](https://www.diabetes.org.uk/guide-to-diabetes/young-adults/type-1-drinking) are also available on the Diabetes UK website (www.diabetes.org.uk (<https://www.diabetes.org.uk/>)).

Advice on smoking and substance misuse

- **Explain about general health problems associated with smoking, particularly the risks of developing vascular complications.**
 - Encourage children and young people with type 1 diabetes not to start smoking.
 - Offer smoking cessation programmes to those who smoke. See the CKS topic on [Smoking cessation \(/topics/smoking-cessation/\)](#) for more information.
- **Explain about the general dangers of substance misuse and the possible effects on blood glucose control.**
 - Information, advice, and support on substance misuse are available on the Talk to Frank website (www.talktofrank.com (<http://www.talktofrank.com/>)).

Basis for recommendation

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How should I manage a child or young person hypoglyc

See the section on [Hypoglycaemia \(/topics/insulin-therapy-in-type-1-diabetes/management/insulin-therapy-type-1-diabetes/#managing-hypoglycaemia\)](#) in the CKS topic on [Insulin therapy in type 1 diabetes \(/topics/insulin-therapy-in-type-1-diabetes/\)](#) for information on how to manage hypoglycaemia.

How should I manage a child or young person with diabetic ketoacidosis?

- If diabetic ketoacidosis (DKA) is **suspected** ([/topics/diabetes-type-1/diagnosis/when-to-suspect-diabetic-ketoacidosis/](#)) in a child or young person, arrange immediate admission to a hospital with acute paediatric facilities, for confirmation of the diagnosis and emergency treatment with intravenous insulin and fluids.
- **After a child or young person has recovered from an episode of DKA:**
 - Discuss with them and/or their family/carers the [factors \(/topics/diabetes-type-1/diagnosis/when-to-suspect-diabetic-ketoacidosis/\)](#) that may have led to the episode.
 - If DKA is recurrent, consider the possibility of non-adherence to treatment or intentional insulin omission in order to lose weight ([diabulimia \(/topics/diabetes-](#)

[type-1/background-information/complications/\)\).](#)

- Assess the emotional and psychological wellbeing of children and young people with type 1 diabetes who present with frequent episodes of DKA.
- Give general information on DKA, including information on how to reduce the risk of future episodes.
 - Encourage the child or young person and/or their family/carers to attend diabetes clinic regularly (4 times per year) as this is associated with good glucose control.
 - Information on DKA for people with type 1 diabetes and their family/carers is available on the Diabetes UK website (www.diabetes.org.uk (https://www.diabetes.org.uk/guide-to-diabetes/complications/diabetic_ketoacidosis/)).
- Ensure that the child or young person and/or their family/carer have appropriate information on how to manage diabetes during a period of illness ('[sick-day rules](#)' ([/topics/diabetes-type-1/management/management-children-young-people/#sick-day-rules](https://www.diabetes.org.uk/topics/diabetes-type-1/management/management-children-young-people/#sick-day-rules))).

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How should I manage a child or young person with type 1 diabetes during a period of illness?

- **Consider the need for admission or specialist advice.** The decision will depend on clinical judgement, taking into account the person's age, any underlying illness or complications, and the presence of hyperglycaemia (random plasma glucose more than 11 mmol/L) and ketosis.
 - **Arrange immediate hospital admission if:**
 - There is an immediate risk of [diabetic ketoacidosis](#) ([/topics/diabetes-type-1/diagnosis/when-to-suspect-diabetic-ketoacidosis/](https://www.diabetes.org.uk/topics/diabetes-type-1/diagnosis/when-to-suspect-diabetic-ketoacidosis/)) (DKA).
 - There is moderate ketonaemia (1.5–2.9 mmol/L) with or without hyperglycaemia and the person cannot eat or drink, as there is a risk of DKA.
 - The child or young person does not improve rapidly with insulin treatment.
 - **Consider arranging hospital admission or seeking urgent specialist advice (depending on clinical judgement) if:**
 - The underlying condition is unclear.
 - The child or young person is dehydrated or at risk of dehydration.
 - Vomiting persists beyond 2 hours (particularly in young children).
 - The child is:
 - Becoming exhausted, confused, or dehydrated; is hyperventilating; or has severe abdominal pain.
 - Younger than 3 years of age or has a co-existing medical condition.
 - The child or young person and/or their family/carers are unable to keep the blood glucose level above 3.5 mmol/L.
 - The child or young person is on continuous subcutaneous insulin pump therapy.
 - The child or young person and/or their family/carers are exhausted, for example due to repeated night-time waking.
 - Language problems make it difficult to communicate with the family.
- **If the child or young person can be managed in primary care:**
 - Assess and manage the intercurrent illness.

- Assess how well blood glucose has been monitored and managed with diet and insulin.
- Warn the person and/or their family/carers that illness may affect blood glucose control:
 - Some illnesses (especially if associated with fever) can raise blood glucose levels.
 - In contrast, illnesses associated with vomiting and diarrhoea (such as gastroenteritis) can lower blood glucose levels, possibly leading to hypoglycaemia (blood glucose levels less than 3.5 mmol/L).
- **Ensure that the child or young person has:**
 - Written 'sick-day rules' (</topics/diabetes-type-1/management/management-children-young-people/#sick-day-rules>) and contact details of the healthcare team.
 - Easily digestible food and sugary drinks (to provide energy and to prevent further ketosis).
 - Oral rehydration salt sachets (to prevent dehydration, particularly in young children).
 - Additional supplies of insulin.
 - Glucose tablets or oral gel (to prevent hypoglycaemia).
 - A glucagon kit (if appropriate).

What are the 'sick-day rules' for children and young people with type 1

- **During a period of illness that does not require admission, remind the child or young person and/or their family/carers to adhere to the 'sick-day rules' that should have been provided by their diabetes team. They should:**
 - **Never stop or omit insulin.**
 - The dose of insulin may need to be altered during periods of illness; they should seek advice from their diabetes team if they are unsure of how to adjust insulin doses.
 - **Check blood glucose more frequently, for example every 1–2 hours including through the night.**
 - The insulin dose should be titrated according to the blood glucose results and the written 'sick-day rules'.
 - **Check for blood ketone levels regularly, for example every 3–4 hours including through the night, and sometimes every 1–2 hours depending on results.**
 - Blood ketones should be checked regardless of blood glucose levels. Rarely, ketone levels may be elevated even if blood glucose levels are normal, for example in gastroenteritis
 - If the blood ketone level is elevated (greater than 3 mmol/L), they should contact the GP or diabetes care team immediately.
 - Advise that the blood ketone testing strips should not be used after the expiry date (as they will not provide a reliable result).
 - **Maintain their normal meal pattern (where possible) if appetite is reduced.**
 - Their normal meals could be replaced with carbohydrate-containing drinks (such as milk, milk shakes, fruit juices, and sugary drinks).
 - **Maintain adequate fluid intake to prevent dehydration.**
 - Water or sugar-free fluids are probably most appropriate in the majority of cases where blood glucose levels are normal or high.
 - If blood glucose levels are low, drinks containing glucose are required (or the person should take carbohydrates if possible).
 - Carbonated drinks should be avoided if possible.

- Seek urgent medical advice if they are violently sick, drowsy, or unable to keep fluids down.
 - Intravenous fluids may be required.
- When feeling better, continue to monitor their blood glucose carefully until it returns to normal.
 - This may take some time.
 - Medical advice should be sought if blood glucose remains uncontrolled.
- Further patient information on managing diabetes during a period of illness is available on the Diabetes UK website (www.diabetes.org.uk (<https://www.diabetes.org.uk/guide-to-diabetes/life-with-diabetes/illness>)).

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How should I monitor for complications in children and with type 1 diabetes?

Children and young people with type 1 diabetes should have regular surveillance for the complications of type 1 diabetes. The reason for the surveillance as well as the significance of any abnormal findings should be discussed with the person and/or their family/carers.

Most children and young people with type 1 diabetes are managed in secondary care.

- For a child or young person being managed in primary care, address the following at every review appointment:
 - Measure HbA1c to ensure adequate blood glucose control. Reiterate [HbA1c](/topics/diabetes-type-1/management/management-children-young-people/#target-hba1c) (</topics/diabetes-type-1/management/management-children-young-people/#target-hba1c>) and [glucose self-monitoring](/topics/diabetes-type-1/management/management-children-young-people/#target-glucose-self-monitoring-levels) (</topics/diabetes-type-1/management/management-children-young-people/#target-glucose-self-monitoring-levels>) targets.
 - Offer a review of injection sites and address any injection site problems. See the section on [Managing injection site problems](/topics/insulin-therapy-in-type-1-diabetes/management/insulin-therapy-type-1-diabetes/#managing-injection-site-problems) (</topics/insulin-therapy-in-type-1-diabetes/management/insulin-therapy-type-1-diabetes/#managing-injection-site-problems>) in the CKS topic on [Insulin therapy in type 1 diabetes](/topics/insulin-therapy-in-type-1-diabetes/) (</topics/insulin-therapy-in-type-1-diabetes/>) for more information.
 - Measure height and weight and plot on an appropriate growth chart. Calculate body mass index (BMI).
 - Check for normal growth and/or significant changes in weight because these may reflect changes in blood glucose control.
 - Check the smoking status of the child or young person. If applicable, offer smoking cessation advice. See the CKS topic on [Smoking cessation](/topics/smoking-cessation/) (</topics/smoking-cessation/>) for more information.
 - Assess for psychological problems (such as depression, anxiety, and eating disorders) and social problems (such as alcohol, substance misuse, and relationship problems).
 - Alerting signs include difficulty with self-care and poor blood glucose control (including frequent episodes of [diabetic ketoacidosis](/topics/diabetes-type-1/diagnosis/when-to-suspect-diabetic-ketoacidosis/) (</topics/diabetes-type-1/diagnosis/when-to-suspect-diabetic-ketoacidosis/>)). See the section on [Psychological and social problems](/topics/diabetes-type-1/management/management-children-young-people/#psychological-social-problems) (</topics/diabetes-type-1/management/management-children-young-people/#psychological-social-problems>) for more information.

- **Advise children and young people with type 1 diabetes and/or their family/carers that they should have:**
 - Regular dental examinations, for example every 3–6 months.
 - An eye examination by an optometrist every 2 years until the age of 12 years when they will be eligible for the annual [diabetes eye screening \(/topics/diabetes-type-1/management/management-children-young-people/#eye-disease\)](/topics/diabetes-type-1/management/management-children-young-people/#eye-disease).
 - Their feet assessed for [diabetic foot problems \(/topics/diabetes-type-1/management/management-children-young-people/#diabetic-foot-problems\)](/topics/diabetes-type-1/management/management-children-young-people/#diabetic-foot-problems).
- **Once a year, offer monitoring for:**
 - Moderately increased albuminuria (albumin:creatinine ratio [ACR] 3–30 mg/mmol) to detect [diabetic kidney disease \(/topics/diabetes-type-1/management/management-children-young-people/#diabetic-kidney-disease\)](/topics/diabetes-type-1/management/management-children-young-people/#diabetic-kidney-disease) (in children aged 12 years and older).
 - Hypertension (in children aged 12 years and older).
 - Thyroid disease — this should be done at diagnosis and annually thereafter until [transfer to adult services \(/topics/diabetes-type-1/management/management-children-young-people/#transition-to-adult-care-services\)](/topics/diabetes-type-1/management/management-children-young-people/#transition-to-adult-care-services).
- **Be aware of the following possible complications and associated conditions:**
 - Juvenile cataracts (rare).
 - Necrobiosis lipoidica (a rare skin condition characterized by shiny, red-brown, or yellowish patches, usually on the shins).
 - Coeliac disease (this should be screened for at diagnosis). See the CKS topic on [Coeliac disease \(/topics/coeliac-disease/\)](/topics/coeliac-disease/) for more information.
 - Addison's disease. See the section on [Diagnosis \(/topics/addisons-disease/diagnosis/\)](/topics/addisons-disease/diagnosis/) in the CKS topic on [Addison's disease \(/topics/addisons-disease/\)](/topics/addisons-disease/) for information on the clinical features of Addison's disease.

Screening for psychological and social problems

Children and young people with type 1 diabetes have a greater risk of emotional and behavioural difficulties than children without diabetes.

- **When managing a child or young person with type 1 diabetes:**
 - **If a behavioural or conduct disorder is suspected**, offer timely access to appropriate mental health professionals. See the CKS topic on [Conduct disorders in children and young people \(/topics/conduct-disorders-in-children-young-people/\)](/topics/conduct-disorders-in-children-young-people/) for more information.
 - **If there are family or relationship difficulties**, offer specific family-based behavioural interventions, such as behavioural family systems therapy, if available.
 - **If there are concerns about psychological wellbeing**, consider referral for a programme of behavioural intervention therapy or behavioural techniques in order to improve:
 - Health-related quality of life, for example counselling or cognitive behavioural therapy (CBT).
 - Blood glucose control in children and young people with high HbA1c levels (HbA1c above 69 mmol/mol [8.5%]).
 - Adherence to diabetes treatment, for example motivational interviewing or multisystemic therapy (an intensive family- and community-based treatment programme that focuses on addressing all environmental systems that impact chronic and violent juvenile offenders).
 - **If anxiety, depression, or an eating disorder (such as anorexia, bulimia, or diabulimia (/topics/diabetes-type-1/background-information/complications/)) is**

suspected, refer promptly to child and adolescent mental health services (CAMHS).

- If there is persistent suboptimal blood glucose control, offer screening for anxiety, depression, and/or eating disorders (which may present with suboptimal overall blood glucose control [both hyperglycaemia and hypoglycaemia] and/or symptoms of gastroparesis).
- See the CKS topics on [Generalized anxiety disorder \(/topics/generalized-anxiety-disorder/\)](/topics/generalized-anxiety-disorder/), [Depression in children \(/topics/depression-in-children/\)](/topics/depression-in-children/), and [Eating disorders \(/topics/eating-disorders/\)](/topics/eating-disorders/) for more information on managing these conditions.

Screening for eye disease

- **Advise children and young people with type 1 diabetes and/or their family/carers that** they should have an eye examination by an optometrist every 2 years until the age of 12 years when they will be eligible for the annual [NHS diabetic eye screening \(DES\) programme \(https://www.gov.uk/guidance/diabetic-eye-screening-programme-overview\)](https://www.gov.uk/guidance/diabetic-eye-screening-programme-overview).
 - The NHS DES is offered once a year to anyone with diabetes who is aged 12 years or over.
 - The programme aims to reduce the risk of sight loss among people with diabetes by the early detection and treatment, if needed, of sight-threatening retinopathy.
- **Explain to the child or young person and/or their family/carers that:**
 - Monitoring for diabetic retinopathy begins at 12 years because diabetic retinopathy that needs treatment is extremely rare in children and young people younger than 12 years of age.
 - Background retinopathy is often found through monitoring, and improving blood glucose control will reduce the risk of this progressing to significant diabetic retinopathy.
 - Annual monitoring from 12 years of age is important because if significant diabetic retinopathy is found, early treatment will improve the outcome.
- **Refer children to the local diabetic screening programme prior to their twelfth birthday to ensure that they are seen from 12 years of age.**

Screening for diabetic kidney disease

From the age of 12 years, children with type 1 diabetes (with or without detected nephropathy) should receive annual screening for diabetic nephropathy.

- **Use the first urine sample of the day ('early morning urine') to measure the albumin:creatinine ratio (ACR).** If this is not possible, use a random sample but be aware that this is associated with an increased risk of false positive results.
 - If the initial ACR is above 3 mg/mmol but below 30 mg/mmol, confirm the result by repeating the test on two further occasions using first urine samples of the day before starting further investigations and therapy.
 - If the initial ACR is 30 mg/mmol or more (proteinuria), arrange further investigations. See the CKS topic on [Chronic kidney disease \(/topics/chronic-kidney-disease/\)](/topics/chronic-kidney-disease/) for more information.
- **Explain to the child or young person and/or their family/carers that:**
 - Monitoring for moderately increased albuminuria (ACR 3–30 mg/mmol) to detect diabetic kidney disease begins at 12 years of age because diabetic kidney disease is extremely rare in younger children.
 - Using the first urine sample of the day ('early morning urine') to screen for moderately increased albuminuria reduces the risk of false positive results.

- If moderately increased albuminuria is detected, improving blood glucose control will reduce the risk of this progressing to significant diabetic kidney disease.
- Annual monitoring from 12 years is important because if diabetic kidney disease is found, early treatment will improve the outcome.

Screening for diabetic foot problems

- **For children with diabetes who are younger than 12 years of age**, give basic foot care advice.
- **For young people with diabetes aged between 12–17 years**, ensure that the paediatric care team or the transitional care team assesses their feet as part of their annual assessment, and provide information on foot care.
 - If a diabetic foot problem is found or suspected, the young person should be referred to an appropriate specialist.

Basis for recommendation

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How should I support the transition from paediatric to services?

- **Explain to young people with type 1 diabetes who are preparing for transition to adult services that some aspects of diabetes care will change at transition**, for example clinic time and location, diabetes specialist team, and self-monitoring blood glucose targets.
- Detailed information on the transition to adult clinics for young people with type 1 diabetes and their family/carers is available on the Diabetes UK website (www.diabetes.org.uk (<https://www.diabetes.org.uk/Guide-to-diabetes/Teens/Me-and-my-diabetes/Transition-to-adult-clinics/>)).

Basis for recommendation

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